IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-150

1. Topic (12 words max)

Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Students in under-resourced schools, particularly in rural Sri Lanka, often have limited access to laboratory facilities, equipment, and guided practical instruction, which weakens conceptual understanding and reduces opportunities for hands-on physics learning [1], [2]. Virtual laboratories are increasingly recognized as effective alternatives because they can improve conceptual understanding, engagement, and access to experimentation without requiring full physical laboratory infrastructure [1], [3]. However, existing virtual-lab research is still fragmented, with many systems focused mainly on simulations or activity logs rather than integrated school-level learning support [3].

At the same time, gamification and AI have shown strong potential to improve science learning motivation, personalization, and learner support [4], [6], [7]. Yet these functions are often studied as separate solutions. Many existing systems provide either simulations, quiz-based assessment, gamified activities, AI tutoring, or content retrieval, but rarely combine them into a single, curriculum-aligned mobile platform for school physics [3], [4], [5]. Furthermore, intelligent educational retrieval systems exist, but they are usually designed as standalone architectures rather than being tightly connected to lesson-level learning activities and student-facing support [5].

Another key limitation is that many digital learning environments do not sufficiently integrate authenticated user management with lesson-based quiz evaluation, practical performance tracking, and progress visualization in one unified student profile [3], [6]. This is particularly important for Grade 6–11 learners, where structured lesson navigation, age-appropriate AI explanations, and clear progress monitoring can significantly support self-learning. Therefore, a clear research gap exists in the design of a **mobile, low-resource, curriculum-aligned virtual physics learning system** that integrates interactive experiments, gamified lesson activities, intelligent content retrieval, grade-adaptive AI support, and lesson-wise learning performance monitoring for rural Sri Lankan students [2], [4], [6], [7].

References

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"Investigating the Effects of AI Chatbots on Students' Learning Outcomes and Motivation."
Educational Technology & Society, vol. 22, no. 2, pp. 1–13, 2019.

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[7] Artificial Intelligence in Education (Systematic Review)

X. Zawacki-Richter, V. I. Marín, M. Bond, and F. Gouverneur,
"Systematic Review of Research on Artificial Intelligence Applications in Higher Education."
International Journal of Educational Technology in Higher Education, 2019.

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6. Brief description of Existing Research and Systems (300 words max)

Recent studies highlight the increasing use of **virtual laboratories, gamified learning systems, and artificial intelligence technologies** in science education. Platforms such as PhET simulations and Labster provide interactive experiment environments where students can visualize scientific concepts and perform virtual laboratory activities. Research has shown that such simulations can improve conceptual understanding and increase student engagement compared to traditional lecture-based teaching.

In addition, digital learning systems often incorporate quizzes and gamified elements to motivate students and reinforce learning. Gamification techniques such as points, levels, and interactive activities have been shown to increase learner motivation and participation in science subjects. However, many existing gamified learning systems focus mainly on entertainment aspects rather than integrating them with structured lesson-based educational objectives.

Artificial intelligence technologies are also increasingly used in education through intelligent tutoring systems and AI chatbots. These tools provide automated explanations, personalized feedback, and guidance to students while they study. Despite these advancements, many AI educational tools primarily focus on answering questions rather than supporting structured learning navigation across curriculum topics.

Another limitation in current systems is the lack of integrated **learning performance monitoring mechanisms** that combine quiz evaluation and practical activity performance within a single student learning profile. Most platforms provide either experiment interaction logs or quiz scores separately, without offering a unified system that tracks lesson-wise conceptual understanding.

Furthermore, many existing educational platforms are designed for environments with strong technological infrastructure and stable internet connectivity. These systems may not be suitable for rural learning environments where students primarily rely on mobile devices.

Therefore, there remains a need for an integrated mobile learning system that combines **interactive physics experiments, gamified learning activities, AI-assisted concept explanation, and lesson-based learning progress monitoring** while remaining lightweight and accessible for rural students.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed system is a mobile-based virtual physics laboratory application developed to support Grade 6–11 students studying physics in rural schools in Sri Lanka. Many schools in these regions lack access to proper laboratory facilities, equipment, and trained instructors, which limits opportunities for practical experimentation. The proposed solution aims to overcome these challenges by transforming smartphones into interactive virtual physics laboratories that enable students to learn physics concepts through experimentation, guided explanations, and structured digital learning activities.

The application provides interactive physics experiments designed to simulate real laboratory activities in a simplified digital environment. Students can interact with virtual objects and perform experiments related to physics concepts such as motion, force, pressure, electricity, and measurement. For example, students may interact with a virtual object such as a moving car, measure the distance it travels, and observe how physical quantities change during the experiment. These interactions allow students to visualize abstract physics principles and understand the relationship between theoretical formulas and real-world behavior.

To reinforce conceptual understanding, the system includes lesson-based quizzes aligned with the national physics syllabus. Each lesson contains multiple quiz questions designed to evaluate the student's understanding of the topic after completing the lesson or experiment. These quizzes provide immediate feedback to help students identify mistakes and improve their comprehension of the subject matter.

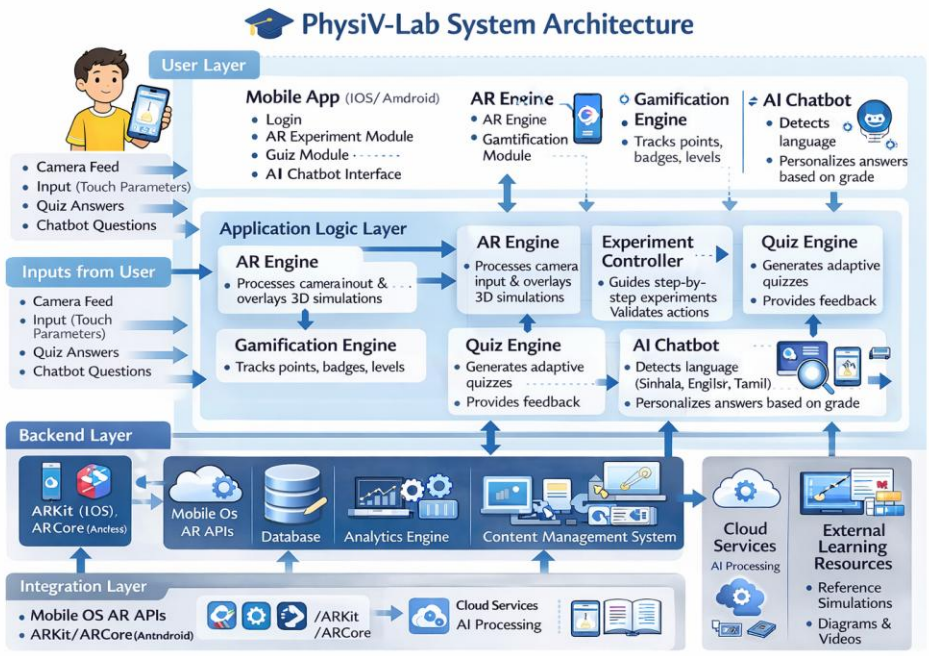
The application also integrates gamified learning activities that convert physics concepts into engaging educational games. Each lesson may contain a simple 2D interactive game designed to reinforce the concepts taught in the lesson. For instance, a physics concept related to motion or distance can be represented through a small game where students interact with objects and observe the results of their actions. These activities increase student engagement and motivation while supporting concept reinforcement.

An important feature of the system is the AI-powered chatbot, which assists students by answering questions related to physics lessons. The chatbot provides explanations adapted to the student's grade level, ensuring that the responses are understandable for students between Grades 6 and 11. The chatbot can explain physics formulas, assist with numerical problem-solving, and guide students through experiment procedures.

The system also includes a learning performance monitoring module that records student activity within the application. Quiz results and experiment performance are stored in the student profile, allowing the system to track lesson-based learning progress over time. This information is presented through simple progress indicators that help students visualize their improvement across different physics topics.

Additionally, the application includes intelligent content retrieval functionality that enables students to search for physics concepts using keywords. The system retrieves relevant explanations and learning materials related to the lesson topic, allowing students to quickly access additional information when needed.

By combining **virtual experiments, gamified learning, AI assistance, intelligent content retrieval, and performance monitoring**, the proposed solution provides an integrated and accessible digital learning environment. This approach helps students in rural areas experience practical physics learning even in the absence of physical laboratory facilities.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful implementation of this project is supported by the availability of multidisciplinary expertise, established research knowledge, and accessible data sources. The proposed system integrates concepts from physics education, artificial intelligence, and interactive digital learning technologies, all of which are supported by existing academic and technical resources. These areas of expertise provide a strong foundation for designing an effective mobile-based virtual physics laboratory application that supports conceptual learning, practical experimentation, and intelligent assistance for students.

Domain Expertise:
The project benefits from domain knowledge in physics education and curriculum design. The learning content, experiments, quizzes, and educational activities are aligned with the Sri Lankan national physics curriculum for Grades 6–11. This ensures that the digital experiments and learning activities correspond with the topics students study in school. Understanding the subject domain allows the system to represent physics concepts such as motion, force, pressure, electricity, and measurement in a simplified and interactive manner that is appropriate for school-level learners.

Technical Knowledge and Tools:
The system development relies on technical knowledge in mobile application development, artificial intelligence integration, and interactive digital simulation design. Mobile development frameworks allow the application to run efficiently on Android devices commonly used by students. Interactive 2D simulation tools are used to design virtual practical experiments and physics-based educational games where students can interact with objects and observe measurable results such as distance, speed, or motion. In addition, AI technologies are used to implement a chatbot capable of providing explanations and guidance related to physics lessons. These technologies together enable the creation of an engaging and accessible digital learning environment.

Data Availability:
The development of the learning content relies on widely available educational resources such as official student textbooks, teacher guides, and physics lesson materials used in Sri Lankan schools. These resources provide the theoretical explanations, formulas, and experiment structures required to design the virtual laboratory activities and quizzes within the application. Using curriculum-based learning materials ensures that the content presented in the system is accurate, relevant, and aligned with the topics students are expected to study at their grade level.

Feasibility:

The proposed system is technically feasible because it uses lightweight mobile technologies and simple interactive simulations that can run on commonly available smartphones. The application does not require specialized laboratory equipment or high-performance computing resources. Furthermore, many of the required development tools, simulation frameworks, and AI integration technologies are readily available and well documented. The modular design of the system allows different components—such as experiments, quizzes, chatbot assistance, and progress monitoring—to be developed and integrated progressively. This approach ensures that the system can be implemented within the available project timeframe while remaining practical for use in rural educational environments.

9. Objectives and Novelty

Main Objective

To design and implement a **mobile-based virtual physics laboratory application** that enables Grade 6–11 students in rural Sri Lanka to learn physics concepts through interactive experiments, gamified activities, AI-assisted explanations, and lesson-based learning progress monitoring.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Sachintha Nirmal S.M IT22234834	Design and Development of a Grade-Adaptive RAG-Based AI Chatbot Module and a Structured Learning Workflow Module	• Developed grade-adaptive AI chatbot • Implemented RAG-based knowledge retrieval • Designed backend AI workflow • Created structured learning navigation system • Integrated AI into learning flow • Conducted testing and evaluation	Curriculum- and Workflow-Aware Grade-Adaptive RAG Tutor (CWGA-RAG Tutor)

Arthanayake C .T .H IT22251800	Intelligent Keyword-Based Content Retrieval and Curriculum-Aligned 2D Game Generation Module	• Implement keyword-based physics content retrieval • Generate lesson-based 2D educational games • Aligning games with curriculum topics	Grade-Responsive Syllabus-Aware Intelligent Search and Adaptive 2D Gameplay-Level Scoping Framework
Wijerathna C .D .W IT22123268	Adaptive User Management and Learning Performance Evaluation Module	• Implement student account registration and login system within overall user management • Track quiz results and practical experiment performance • Monitor lesson-based learning progress	Integrates quiz evaluation, practical experiment performance, and lesson-based learning progress monitoring within a unified student learning analytics module for mobile virtual physics education.
Vidurangi H. I .D .G IT22143990	Interactive Physics Experimentation and Learning Module	• Develop interactive physics simulations that generate measurable results • Implement virtual experiments using digital laboratory tools	Measurement-driven interactive simulations that automatically connect student actions with real-time physics calculations and conceptual explanations.

		• Design step-by-step physics calculation explanations based on experiment results	

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr	Indunil	Dalawatte	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				